Solution Requirements (Functional & Non-functional)

Date	12-06-2026
Team ID	
Project Name	**Voice Based Concept Understanding Analyser**
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication	User Registration through Form User Login using Email & Password Secure Logout
FR-2	Voice Recording & Upload	Record Voice Response Upload Audio File (.wav/.mp3) Validate Audio Format
FR-3	Speech Processing	Convert Speech to Text using Whisper Preprocess Audio Extract Speech Features
FR-4	Concept Understanding Evaluation	Compare Student Answer with Reference Answer using Sentence-BERT, Calculate Semantic Similarity Score, Evaluate Concept Accuracy
FR-5	Speech Quality Analysis	Analyse Speaking Rate, Detect Pauses & Fluency, Evaluate Communication Quality
FR-6	AI Feedback Generation	Generate Personalised Feedback using Gemini AI, Suggest Improvements, Highlight Missing Concepts

Non-functional Requirements:

The following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The application should provide an intuitive, responsive, and user-friendly interface that allows students to complete evaluations with minimal training
NFR-2	**Security**	User data, audio recordings, and reports should be securely stored. API keys and sensitive information must be protected, and secure authentication should be implemented.
NFR-3	**Reliability**	The system should consistently produce accurate evaluations and continue functioning even if external AI services experience temporary failures.
NFR-4	**Performance**	Voice processing and AI evaluation should complete within a reasonable response time (typically under 10–15 seconds for standard audio inputs).
NFR-5	**Availability**	The application should be accessible whenever users need it and support continuous operation with minimal downtime
NFR-6	**Scalability**	The architecture should support increasing numbers of users and future deployment on cloud platforms without significant performance degradation.